
EDF Battery Design Project

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Abstract

This project presents an engineering-oriented modeling framework linking experimental lithium-ion battery cell data to electric vehicle range estimation. A 0th-order equivalent circuit model is identified at cell level from laboratory characterization tests and scaled to a full battery pack. The battery model is coupled with a longitudinal electric vehicle model to simulate energy consumption over a WLTP Class 3 driving cycle.

The proposed approach predicts a driving range of 681 km for a fresh battery, which underestimates the 740 km reference range reported for a Tesla Model S by approximately 8.0%. Given the simplicity of the adopted models, this level of agreement is considered satisfactory. The framework also enables the evaluation of ageing effects, showing a progressive reduction in range as battery state of health decreases.

While the model neglects battery polarization dynamics, temperature effects, road slope, and assumes constant drivetrain efficiencies and auxiliary loads, it remains well suited for battery pack sizing and comparative range analysis. The proposed methodology provides a consistent and transparent baseline for more detailed future modeling efforts.

Part I

Context and Objectives

The rapid development of electric vehicles (EVs) raises major challenges related to battery sizing, energy efficiency, and performance degradation over time. Accurate battery and vehicle models are therefore essential to support engineering decisions during early design phases.

This project is part of the *Data and Modelling Weeks* organized by CentraleSupélec in collaboration with EDF R&D. The objective of this project is to estimate the driving range of the Tesla Model S subjected to the WLTP test cycle. To achieve this, a longitudinal electric vehicle model is first used to compute the mechanical power required at the wheels over the WLTP speed profile. This mechanical power demand is then

converted into electrical power which is supplied to a battery pack model. This pack model computes the battery current as a function of time, which is used to derive the evolution of the state of charge and thus eventually the vehicle driving range.

The work also focuses on analysing the impact of certain battery and EV parameters on the range of the vehicle.

Part II

Materials and Methods

1 Electric Vehicle Model

In this section, we present the electric vehicle (EV) longitudinal model developed to compute the electrical power required at the battery terminals as a function of time. The model uses the WLTP Class 3 speed profile as input and is progressively refined to improve physical realism while preserving computational simplicity.

To design the model, the relevant input parameters were first identified, followed by the definition of a set of simplifying assumptions regarding vehicle dynamics and operating conditions.

The EV model relies on the following simplifying assumptions:

- The vehicle motion is purely longitudinal; lateral and vertical dynamics are neglected.
- Road slope is assumed to be zero throughout the driving cycle.
- Rolling resistance coefficient C_{rr} is constant.
- Aerodynamic drag coefficient C_d and frontal area A_f are constant.
- Traction and regenerative efficiencies are constant.
- During deceleration the negative power is assumed to come entirely from regenerative braking.
- Auxiliary power consumption P_{aux} is constant.
- Tire slip and drivetrain transient effects are neglected.

The main input parameters of the wheel power model are summarized below.

Parameter	Unit	Description
m	kg	Vehicle mass including passengers and payload
C_{rr}	–	Rolling resistance coefficient of the tires
C_d	–	Aerodynamic drag coefficient
A_f	m ²	Vehicle frontal area exposed to airflow
η_{trac}	–	Traction efficiency (motor, inverter, drivetrain)
η_{regen}	–	Regenerative braking efficiency
P_{aux}	W	Constant auxiliary electrical power consumption
t	s	Time vector of the driving cycle
v	km/h	Vehicle speed from the driving cycle

Table 1: Input parameters of the wheel power model

The wheel power model was developed through three successive iterations. Each iteration introduces additional physical effects in order to improve the accuracy of the estimated battery power demand, while maintaining a modeling approach suitable for system-level simulations.

In the first iteration, a basic longitudinal vehicle model is considered. Only the fundamental resistive forces acting on the vehicle are taken into account, namely inertial forces, rolling resistance, and aerodynamic drag. The traction force required to follow the driving cycle is expressed as

$$F(t) = m \frac{dv}{dt} + mgC_{rr} + \frac{1}{2}\rho C_d A_f v^2, \quad (1)$$

leading to the mechanical power at the wheels

$$P_{\text{mech}}(t) = F(t) v(t). \quad (2)$$

This first iteration provides a physically meaningful baseline model and allows estimation of the mechanical power required at the wheels. However, all drivetrain losses are neglected, meaning that the computed power does not represent the actual electrical demand seen by the battery. As a result, this formulation tends to underestimate energy consumption.

In the second iteration, drivetrain efficiency is introduced in order to bridge the gap between mechanical wheel power and electrical battery power. A constant traction efficiency η_{trac} is applied during propulsion phases, yielding

$$P_{\text{bat}}(t) = \frac{P_{\text{mech}}(t)}{\eta_{\text{trac}}}, \quad P_{\text{mech}} \geq 0. \quad (3)$$

This improvement allows the model to account for losses in the motor, inverter, and transmission, significantly improving the realism of the battery power estimation during driving. Nevertheless, braking phases

are still treated in a purely mechanical sense and auxiliary electrical loads remain neglected.

In the third iteration, the model is further extended to include both regenerative braking and auxiliary electrical consumption. When the mechanical power becomes negative, a regenerative efficiency η_{regen} is applied to represent partial energy recovery:

$$P_{\text{bat}}(t) = P_{\text{mech}}(t) \eta_{\text{regen}}, \quad P_{\text{mech}} < 0. \quad (4)$$

In this model, we assume that during deceleration the braking torque is provided by the electric machine only. In simpler terms, Whenever the vehicle slows down, the wheels are slowed down only by the electric motor, not by the mechanical brakes. This assumption is justified because the WLTP cycle contains smooth and moderate decelerations, for which modern electric vehicles—such as the Tesla Model S—prioritize regenerative braking. Friction brakes are mainly used during emergency braking, at very low speeds, or when regenerative limits are reached, which are outside the scope of this simplified model. Losses during regeneration are accounted for through a regenerative efficiency coefficient

In addition, a constant auxiliary power demand P_{aux} is added to represent onboard electrical loads such as HVAC systems, control units, and infotainment:

$$P_{\text{bat}}(t) = \begin{cases} \frac{P_{\text{mech}}(t)}{\eta_{\text{trac}}} + P_{\text{aux}}, & P_{\text{mech}} \geq 0, \\ P_{\text{mech}}(t) \eta_{\text{regen}} + P_{\text{aux}}, & P_{\text{mech}} < 0. \end{cases} \quad (5)$$

This final formulation captures the dominant contributors to battery power demand over a driving cycle and enables realistic energy consumption and range estimation.

Traction and regenerative braking efficiencies are set using constant, cycle-averaged values representative of a modern battery electric vehicle drivetrain, such as that of a Tesla Model S. The traction efficiency is set to $\eta_{\text{trac}} = 0.90$, which reflects the combined average efficiency of the electric motor, power inverter, and single-speed transmission over a standard driving cycle. This value is consistent with reported drivetrain efficiencies for contemporary electric vehicles, which typically range between 0.88 and 0.94 when averaged over WLTP-like conditions.¹

The regenerative braking efficiency is assumed to be $\eta_{\text{regen}} = 0.70$. This value represents the fraction of mechanical braking energy that is effectively recovered and stored in the battery, accounting for losses in the electric machine, power electronics, battery charge acceptance, and brake blending effects. Cycle-averaged regenerative efficiencies reported in the literature commonly fall within the 0.6–0.8 range, making the chosen value a realistic and conservative assumption for WLTP.

¹G. L. Plett, *Battery Management Systems, Volume II: Equivalent-Circuit Methods*, Artech House, 2015.

based simulations.²

The auxiliary electrical power consumption is modeled as a constant load of $P_{\text{aux}} = 800$ W. This term aggregates the power demand of non-traction systems such as vehicle control units, sensors, infotainment, cooling pumps, lighting, and low-voltage power electronics. The selected value lies within the commonly reported range of 500–1000 W for mid- to high-end electric vehicles operating under nominal conditions without aggressive HVAC usage, and is consistent with system-level EV modeling practices.³

More advanced models could further incorporate speed-dependent efficiencies, road slope, thermal effects, or higher-order electrical dynamics. However, such additions would significantly increase model complexity and parameter identification effort, while providing limited benefit for the objectives of this project. The third iteration was therefore selected as an appropriate compromise between model accuracy and computational simplicity, well suited for system-level battery sizing and ageing studies.

2 Single Cell Electrical Model

Before scaling the battery behavior to pack level, it is necessary to establish a reliable electrical model of a single cell. The objective of this section is to introduce the chosen equivalent circuit model and describe how its parameters are identified from experimental pulse data.

2.1 State of Charge (SOC) and State of Health (SOH) calculation. The state of charge (SOC) represents the relative amount of charge remaining in the battery compared to its usable capacity. It is defined as the ratio between the remaining charge and the cell capacity, and is expressed as a percentage.

In this project, SOC is computed using the coulomb counting method. Starting from a known initial SOC value $\text{SOC}(t_0)$, the SOC evolution over time is given by

$$\text{SOC}(t) = \text{SOC}(t_0) + \frac{1}{Q_{\text{cell}}} \int_{t_0}^t I_{\text{cell}}(\tau) d\tau, \quad (6)$$

where Q_{cell} is the cell capacity expressed in ampere-hours and I_{cell} is defined as negative during discharge.

The state of health (SOH) quantifies the ageing state of a battery cell by comparing its current usable capacity to its capacity when new. SOH is defined as

$$\text{SOH} = \frac{Q_{\text{current}}}{Q_{\text{new}}}, \quad (7)$$

²R. Xiong *et al.*, “Critical Review on the Battery State of Charge Estimation Methods for Electric Vehicles,” *IEEE Access*, vol. 6, pp. 1832–1843, 2018.

³R. Xu, *Lithium-ion Battery Modeling and State of Charge Estimation*, Master’s Thesis, KTH Royal Institute of Technology, Stockholm, Sweden, 2023.

where Q_{current} is obtained by integrating the discharge current over a complete discharge test:

$$Q_{\text{aged}} = \int I_{\text{cell}}(t) dt. \quad (8)$$

2.2 0th-order Thevenin (Rint) equivalent circuit model. The simplest electrical representation of a lithium-ion cell is the 0th-order Thevenin model, also referred to as the *Rint* model. This model captures the dominant static electrical behavior of the cell while remaining computationally lightweight and easy to parameterize.

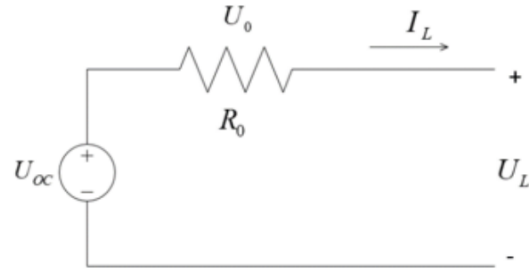


Figure 1: 0th-order Thevenin (*Rint*) equivalent circuit model of a lithium-ion cell

The model consists of:

- an ideal voltage source representing the open-circuit voltage $\text{OCV}(\text{SOC})$,
- a single internal ohmic resistance $R_0(\text{SOC})$ accounting for instantaneous voltage losses.

Under a load current I_{cell} , the terminal voltage of the cell is given by

$$V_{\text{cell}} = \text{OCV}(\text{SOC}) + I_{\text{cell}} R_0(\text{SOC}). \quad (9)$$

The + sign is used here as current is already negative during discharge.

2.3 Extraction of the ohmic resistance from current pulse tests. The parameter R_0 is identified experimentally using current pulse tests.

Starting from a rest condition ($I = 0$), the cell is subjected to a current step ΔI . The terminal voltage exhibits an immediate drop ΔV_{ohmic} , followed by slower transient dynamics. Because capacitive and diffusion-related phenomena cannot react instantaneously, this initial voltage drop is attributed exclusively to the ohmic resistance R_0 , which can be computed as

$$R_0 = \frac{\Delta V_{\text{ohmic}}}{\Delta I}. \quad (10)$$

In practice:

- ΔV_{ohmic} is measured immediately before and after the current step,
- ΔI is the amplitude of the applied current pulse.

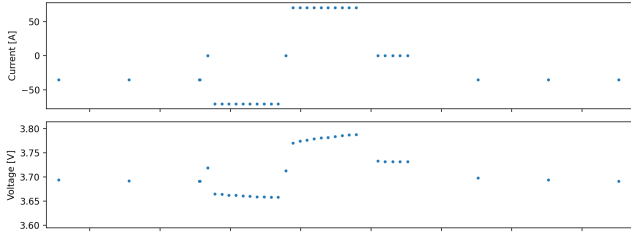


Figure 2: Typical cell voltage response to a current pulse

By repeating this procedure at different SOC and SOH levels, the evolution of $R_0(\text{SOC})$ can be characterized.

2.4 Identification of the OCV–SOC relationship. The open-circuit voltage $OCV(\text{SOC})$ is identified experimentally. When a battery cell is left at rest with zero current, all transient polarization effects vanish and the terminal voltage converges to the open-circuit voltage.

In practice, the $OCV(\text{SOC})$ curve is obtained by applying a sequence of current pulses at different SOC levels, each containing several rest periods. The OCV value at a given SOC is obtained by averaging the voltage during these rest phases.

2.5 From WLTP power demand to battery current using the 0th-order model. The vehicle model described in Section 2 provides the electrical power requested at the battery terminals as a function of time, denoted $P_{\text{need}}(t)$. The power supplied by the battery therefore satisfies

$$P_{\text{bat}}(t) = -P_{\text{need}}(t). \quad (11)$$

2.5.1 Sign convention. The following sign convention is adopted consistently throughout the model:

- $P_{\text{bat}} < 0$ corresponds to battery discharge,
- $P_{\text{bat}} > 0$ corresponds to battery charge,
- $I_{\text{cell}} < 0$ corresponds to discharge current,
- $I_{\text{cell}} > 0$ corresponds to charge current.

With this convention, the 0th-order Thevenin model is written as:

$$V_{\text{cell}} = OCV(\text{SOC}) + I_{\text{cell}} R_0(\text{SOC}). \quad (12)$$

The electrical power at cell level is:

$$P_{\text{cell}} = V_{\text{cell}} I_{\text{cell}}. \quad (13)$$

2.5.2 Current computation from power demand. Substituting the voltage expression into the power equation yields:

$$P_{\text{cell}} = (OCV(\text{SOC}) + I_{\text{cell}} R_0(\text{SOC})) I_{\text{cell}}. \quad (14)$$

This can be rewritten as a quadratic equation in the current:

$$R_0(\text{SOC}) I_{\text{cell}}^2 + OCV(\text{SOC}) I_{\text{cell}} - P_{\text{cell}} = 0. \quad (15)$$

At each time step the OCV and R_0 is calculated using the SOC based on the relationship calculated using the method described in section 2.3 and 2.4. Then the second-order polynomial (14) is solved using the positive root.

$$I_{\text{step}} = \frac{-OCV + \sqrt{OCV^2 + 4 R_0 P_{\text{cell}}}}{2 R_0} \quad (16)$$

The positive root is selected as if we chose the negative root then at every step $I_{\text{step}} < 0$ regardless of if $P_{\text{cell}} < 0$ or $P_{\text{cell}} > 0$, which doesn't make sense. With this current, we use coulomb counting to recalculate SOC for the next step effectively adding the current of this step to the integration.

$$\text{SOC}_{\text{step}+1} = \text{SOC}_{\text{step}} + \frac{I_{\text{step}} \Delta t}{3600 Q} \quad (17)$$

We can then calculate V_{step} using equation 12. Repeating this for every time step, we obtain the current and voltage profile matching the power profile inputted as well as the SOC degradation due to the power request.

3 Battery pack sizing method and results using a 0th-order Thevenin ECM

3.1 Cell model and identified parameters. Each cell is represented using a 0th-order Thevenin (Rint) equivalent circuit. The terminal voltage is expressed as the open-circuit voltage reduced by an ohmic voltage drop across the internal resistance:

$$V_{\text{cell}} = OCV(\text{SOC}) - I_{\text{cell}} R_0(\text{SOC}). \quad (18)$$

Experimental identification yields the following parameters:

- $OCV(0) \approx 3.1 \text{ V}$, $OCV(1) \approx 4.2 \text{ V}$,
- $R_0(0) \approx 1.5 \text{ m}\Omega$, $R_0(1) \approx 0.8 \text{ m}\Omega$,
- Cell capacity decreases with ageing (SOH) from approximately 66 Ah ($SOH = 1$) to 61.5 Ah ($SOH = 0.928$).

The vehicle model provides the electrical power demand at the battery terminals, with peak values of approximately 62 kW during discharge and –23 kW during regenerative braking. The objective of the sizing procedure is to determine the number of cells in series N_s and in parallel N_p that satisfy voltage, power, and energy requirements.

3.2 Battery pack electrical scaling laws.

3.2.1 Pack voltage scaling. For N_s identical cells connected in series, the pack voltage scales as:

$$V_{\text{pack}}(\text{SOC}) \approx N_s V_{\text{cell}}(\text{SOC}), \quad (19)$$

and the open-circuit voltage becomes:

$$V_{\text{pack,OCV}}(\text{SOC}) = N_s \text{OCV}(\text{SOC}). \quad (20)$$

3.2.2 Pack resistance scaling. When N_s cells are connected in series, their resistances add:

$$R_{\text{series}}(\text{SOC}) = N_s R_0(\text{SOC}). \quad (21)$$

For N_p identical series strings connected in parallel, the equivalent pack resistance is:

$$R_{\text{pack}}(\text{SOC}) = \frac{R_{\text{series}}(\text{SOC})}{N_p} = \frac{N_s}{N_p} R_0(\text{SOC}). \quad (22)$$

The pack terminal voltage under load is then approximated as:

$$V_{\text{pack}}(\text{SOC}) \approx N_s \text{OCV}(\text{SOC}) - I_{\text{pack}} R_{\text{pack}}(\text{SOC}). \quad (23)$$

3.3 Pack energy scaling and cell energy approximation. Electrical energy is written as:

$$E = \int P dt = \int V I dt. \quad (24)$$

Using $I = \frac{dQ}{dt}$, this becomes:

$$E = \int V dQ. \quad (25)$$

For pack sizing purposes, voltage is approximated by a nominal value V_{nom} , yielding:

$$E_{\text{cell}} \approx V_{\text{nom}} \int dQ = V_{\text{nom}} Q_{\text{cell}}. \quad (26)$$

At pack level, the total energy is:

$$E_{\text{pack}} \approx N_s N_p E_{\text{cell}} \approx N_s N_p V_{\text{nom}} Q_{\text{cell}}. \quad (27)$$

3.4 Selection of the number of series cells. A Tesla Model S reference is used to select a realistic voltage level. Public specifications indicate a 400 V-class architecture with a nominal voltage around 407 V. Following this reference:

$$N_s = 110. \quad (28)$$

The resulting pack open-circuit voltage range is:

$$V_{\text{pack,OCV}} \in [110 \cdot 3.1, 110 \cdot 4.2] = [341, 462] \text{ V}, \quad (29)$$

which is consistent with a nominal voltage in the 400 V range.

3.5 Energy sizing and determination of the number of parallel cells. The Tesla Model S has an WLTP-rated range of 740 km and a typical usable battery energy

of approximately 95 kWh, adopted as the target pack energy.

Using $V_{\text{nom}} \approx 3.7$ V and an aged capacity $Q_{\text{cell}} = 60$ Ah:

$$E_{\text{cell}} \approx 3.7 \cdot 60 \approx 222 \text{ Wh}. \quad (30)$$

The required number of cells is:

$$N_{\text{cells}} \approx \frac{95,000}{222} \approx 428. \quad (31)$$

With $N_s = 110$:

$$N_p \approx \frac{428}{110} \approx 3.9 \Rightarrow N_p = 4. \quad (32)$$

The resulting configuration is:

$$N_s N_p = 110 \times 4 = 440 \text{ cells}, \quad (33)$$

with an estimated pack energy:

$$E_{\text{pack}} \approx 440 \cdot 222 \approx 97.7 \text{ kWh}. \quad (34)$$

3.6 Calculation of pack resistance and voltage drop.

The peak discharge power is approximately 62 kW. Using the lowest pack voltage as a conservative estimate, the peak pack current is:

$$I_{\text{pack,peak}} \approx \frac{62,000}{341} \approx 182 \text{ A}. \quad (35)$$

Peak current per cell is therefore $182/4 \approx 45.5$ A. This current value is in the range of currents applied in the test cases. At low SOC, the worst-case internal resistance is $R_0 \approx 1.5$ m Ω , yielding:

$$R_{\text{pack}} \approx \frac{N_s}{N_p} R_0 = \frac{110}{4} \cdot 0.0015 \approx 0.0413 \text{ } \Omega. \quad (36)$$

The corresponding voltage drop at peak current is:

$$\Delta V_{\text{pack}} \approx I_{\text{pack,peak}} R_{\text{pack}} \approx 182 \cdot 0.0413 \approx 7.5 \text{ V}. \quad (37)$$

This voltage drop remains moderate and negligible compared to the nominal pack voltage. Thus our value for N_s remains valid.

3.7 Final proposed pack sizing. Based on voltage level, peak power capability, and energy requirements, the resulting battery pack configuration is:

$$N_s = 110, \quad N_p = 4 \Rightarrow 110s4p \text{ (440 cells)}. \quad (38)$$

3.8 WLTP power demand applied to total pack. At pack level, the total battery power P_{pack} is distributed across the 440 cells:

$$P_{\text{cell}} = \frac{P_{\text{pack}}}{N_s N_p}. \quad (39)$$

Once the cell current I_{cell} is computed at each step, the pack current of one step is obtained using Kirchhoff's current law:

$$I_{\text{pack}} = N_p I_{\text{cell}}. \quad (40)$$

This current is used to update the state of charge via coulomb counting and compute terminal voltage. With this we can determine the SOC degradation of the pack after undergoing one WLTP. Then we can calculate the number of WLTP cycles we would need to fully deplete the battery pack and thus infer the estimate range of the EV of study.

Part III

Results and discussion

The next section presents the results obtained by applying the Materials and Methods described previously. Results are organized following the same workflow, from the electric vehicle model to cell-level and pack-level simulations.

4 Electric vehicle model results

This section presents the results of the electric vehicle dynamics model applied to the WLTP Class 3 driving cycle.

4.1 Mechanical power demand.

The mechanical power required at the wheels is computed from the longitudinal vehicle model.

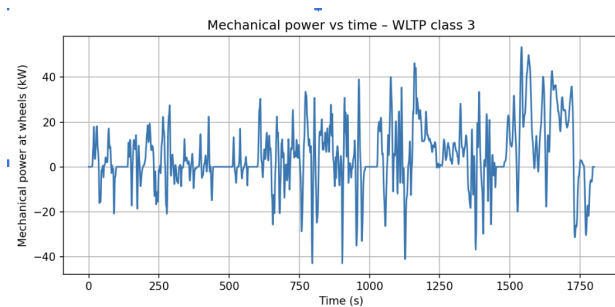


Figure 3: Mechanical power demanded at the wheels over the WLTP Class 3 cycle

4.2 Electrical power demand at the battery.

The corresponding electrical power required at the battery terminals is shown below.

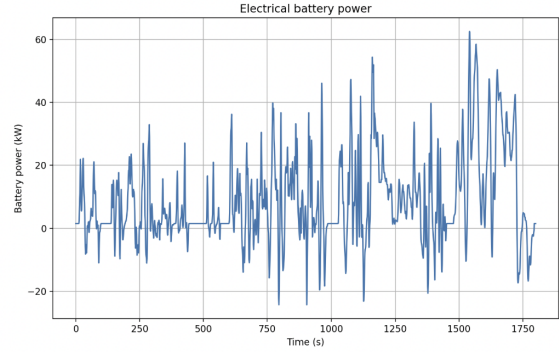


Figure 4: Electrical power demanded at the battery over the WLTP Class 3 cycle

5 Single cell electrical behavior

This section presents results related to the single-cell 0th-order Thevenin equivalent circuit model.

5.1 Ohmic resistance identification.

The instantaneous voltage drop extracted from pulse tests is used to identify the ohmic resistance R_0 as a function of SOC.

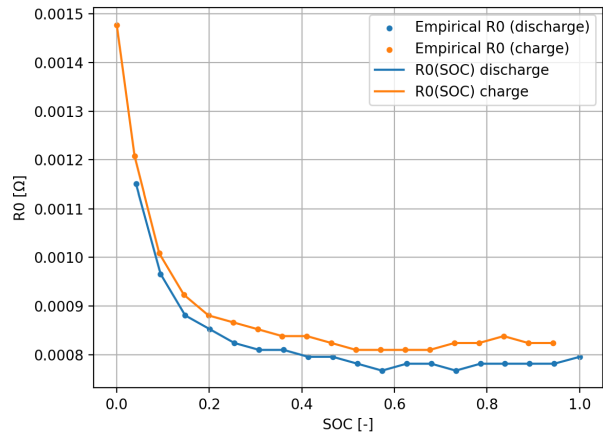


Figure 5: R_0 as a function of SOC

For both charge and discharge conditions, R_0 decreases sharply at low SOC and then stabilizes over a wide mid-SOC range. This behavior is consistent with lithium-ion cell physics, as internal resistance is known to increase at low SOC. The systematic offset between charge and discharge curves indicates a slight asymmetry in internal losses depending on current direction, which is commonly observed in experimental measurements.

The influence of battery ageing on $R_0(SOC)$ is illustrated for different SOH values.

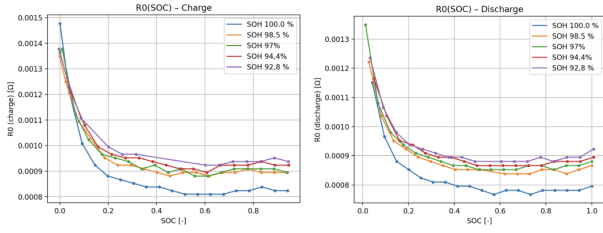


Figure 6: $R_0(SOC)$ for different SOH values

As SOH decreases, R_0 increases across the entire SOC range. The overall shape of the $R_0(SOC)$ curve remains similar for all SOH levels, suggesting that ageing primarily affects the resistance magnitude rather than its functional dependence on SOC.

5.2 Open-circuit voltage characteristic. The open-circuit voltage measured during rest periods is used to identify the $OCV(SOC)$ relationship.

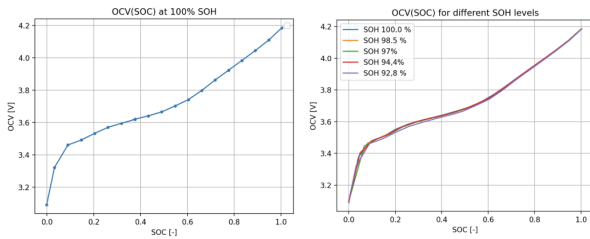


Figure 7: OCV as a function of SOC

The curve shows a relatively steep slope at low SOC, a flatter mid-SOC region, and a steeper rise near full charge. When comparing different SOH levels, the $OCV-SOC$ curves remain very close to one another, indicating that ageing has a limited influence on equilibrium voltage. This confirms that capacity decrease and resistance increase are the dominant ageing effects for the considered cells, rather than shifts in open current voltage.

6 Single cell implementation results

6.1 Order 0 model versus experimental data. The following figure shows the comparison between the measured cell voltage and the voltage predicted by a 0th-order model.

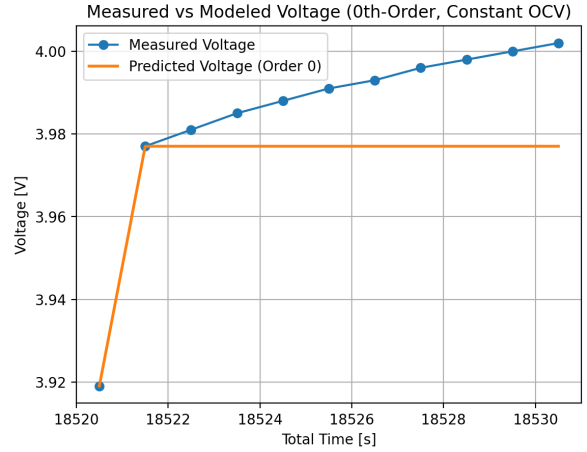


Figure 8: Order 0 model versus experimental data

The 0th-order model captures the initial voltage drop but fails to reproduce the gradual voltage recovery observed experimentally, mostly because of polarization effects.

As a sanity check, the single-cell model is then tested using a scaled down WLTP power profile (Power/1000).

6.2 Cell current evolution. The resulting cell current over time is shown below.

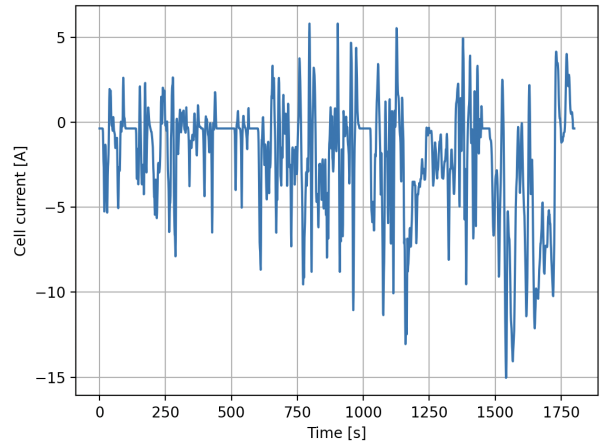


Figure 9: Single-cell current as a function of time

6.3 SOC evolution. The corresponding evolution of SOC during the test is presented for different SOH values.

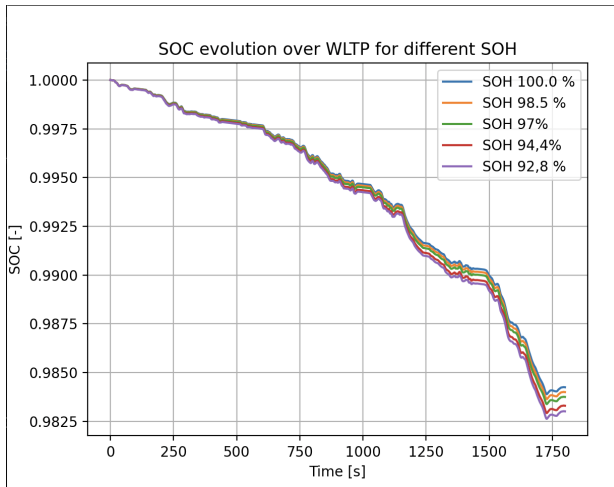


Figure 10: Single-cell SOC evolution over time

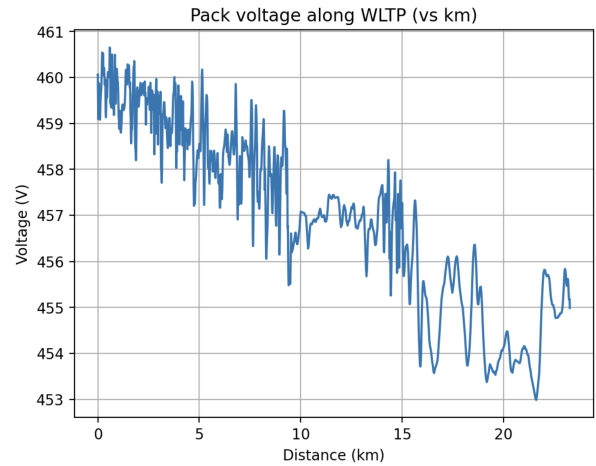


Figure 12: Pack voltage as a function of WLTP distance travelled

7.2 SOC evolution and range estimation.

SOH (%)	100	98.5	97.0	94.4	92.8
Range (km)	681	669	660	643	630

Table 2: Impact of battery state of health on estimated vehicle range

7 Electric vehicle with battery pack simulation results

This section presents results obtained by coupling the battery pack model with the electric vehicle model.

7.1 Pack current and voltage along WLTP. The pack current and voltage are shown as functions of WLTP distance.

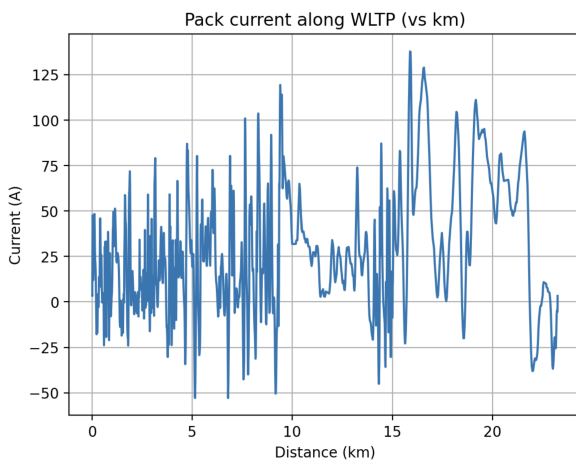


Figure 11: Pack current as a function of WLTP distance travelled

This table quantifies the reduction in vehicle range as a function of battery SOH. As SOH decreases from 1 to 0.928, the estimated range drops from 681 km to 630 km. This reduction is mainly driven by the loss of usable capacity associated with ageing. The relationship between SOH and range appears approximately linear over the investigated range.

The SOC evolution along the WLTP cycle is shown for different SOH values.

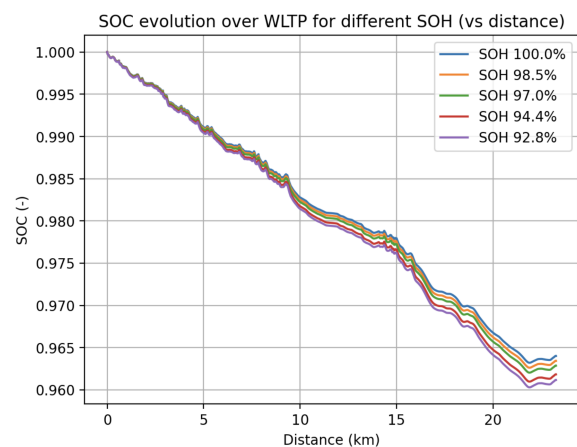


Figure 13: SOC as a function of WLTP distance for different SOH values

Cells with lower SOH experience a slightly faster SOC decrease for a given distance, due to their reduced available capacity. Although the differences remain small

over a single WLTP cycle, they accumulate over longer distances and explain the range reduction observed for aged batteries.

Tesla mentions a range of 744 km for the Tesla Model S.

$$\Delta\% = \frac{|681 - 744|}{744} \times 100 \approx 8.5\% \quad (41)$$

The theoretical range predicted by the model underestimates the reference range by approximately 8.0%. Considering the simplicity of the model, this difference is understandable. Indeed, several limitations arise from the simplifying assumptions. The battery model neglects polarization dynamics and temperature dependence. At vehicle level, motor and drivetrain efficiencies as well as auxiliary power demand are assumed constant. Road slope is also neglected and battery mass may differ from the official battery.

7.3 Sensitivity analysis of the range model.

To understand the discrepancy between the modeled range (681 km) and the reference value (744 km), a one-at-a-time sensitivity analysis was performed using the parameters exposed in the GUI. The objective is not to perfectly calibrate the vehicle, but to identify which assumptions have the strongest influence on the predicted driving range.

A possible source of error is the vehicle mass. Indeed, the commercialized Tesla battery has a different structure than the one in our model, thus the battery mass used in the model may differ from the actual battery mass of the vehicle. There is no official manufacturer data publicly available for the exact battery mass of the Tesla Model S; however, independent teardown and engineering analyses typically report a battery mass in the range of 500–600 kg.

In the present model, the battery pack mass is estimated as 440 kg, assuming a mass of 1 kg per cell for a total of 440 cells. Taking a plausible real battery mass of 550 kg, the effective vehicle mass used in the simulation is corrected as:

$$m_{\text{veh, corr}} = 2167 - (550 - 440) = 2057 \text{ kg.}$$

With this corrected vehicle mass, the simulated driving range becomes 702 km, corresponding to a relative error of 5.6% with respect to the experimental range.

Auxiliary power. Auxiliary power P_{aux} is modeled as a constant electrical load representing HVAC systems, displays, control electronics, sensors, and communication units. Because this load is independent of speed, its energy contribution scales directly with time per kilometer.

For the WLTP Class 3 cycle ($T \approx 1800$ s, $D \approx 23.26$ km), the auxiliary energy per distance is:

$$e_{\text{aux}} = \frac{P_{\text{aux}} T}{3600 D} \approx 0.0215 P_{\text{aux}} \quad [\text{kWh/km}], \quad (42)$$

with P_{aux} expressed in kW.

Using $P_{\text{aux}} = 800$ W, auxiliary loads contribute approximately 1.7 kWh/100km, leading to a predicted range of 681 km at full SOH. Lower auxiliary demand (e.g. 500 W in mild conditions) would further reduce energy consumption and bring the predicted range closer to the official Tesla Model S value of 744 km. This demonstrates that auxiliary power is one of the dominant drivers of range uncertainty in simplified models.

Traction and regenerative efficiencies. Traction efficiency η_{trac} directly affects nearly all propulsion energy and the battery power scales approximately as $1/\eta_{\text{trac}}$. Small variations therefore have a noticeable impact on range. Regenerative efficiency η_{regen} only affects the recoverable fraction of braking energy and has a smaller, secondary influence.

Other parameters. Aerodynamic parameters (C_d , A_f) strongly influence consumption at high speed due to the quadratic dependence on velocity, while rolling resistance C_{rr} introduces a constant opposing force over the entire cycle. Vehicle mass mainly affects acceleration phases and has a comparatively smaller impact on total range for WLTP-type driving.

Parameter	Baseline	Variation	Main effect on range
P_{aux}	800 W	500–1500 W	Strong: constant power draw; major source of range uncertainty
η_{trac}	0.90	± 0.03	Medium–strong: affects propulsion energy ($\sim 1/\eta$ scaling)
η_{regen}	0.70	± 0.10	Moderate: limited to recoverable braking energy
C_d , A_f	fixed	$\pm$ few %	Moderate: aerodynamic losses dominate at high speed
C_{rr}	0.010	± 0.001	Moderate: rolling losses accumulate over distance
m	2167 kg	± 100 kg	Smaller: mainly affects acceleration phases

Table 3: Qualitative sensitivity of the modeled driving range to key parameters

Overall, the predicted range is most sensitive to auxiliary power consumption and traction efficiency. The remaining discrepancy with the official Tesla Model S range is therefore largely attributable to simplifying assumptions on constant auxiliaries and average drivetrain efficiencies, rather than to the battery electrical model itself.

Part IV

Conclusion

This project developed a modeling framework linking experimental battery cell data to electric vehicle range estimation. A 0th-order equivalent circuit model was identified at cell level, scaled to a battery pack, and coupled with a longitudinal vehicle model to simulate performance over a WLTP Class 3 driving cycle.

The model predicts a driving range of 681 km for a fresh battery, compared to the 740 km range reported by Tesla for the Model S, corresponding to a relative difference of approximately 8.5%. Given the simplicity of the adopted models, this level of discrepancy is acceptable and indicates that the main physical phenomena governing energy consumption are reasonably captured.

The observed difference can be attributed to the simplifying assumptions of the model. At cell level, the battery is represented using a 0th-order equivalent circuit that neglects polarization dynamics and temperature effects. At vehicle level, motor and drivetrain efficiencies as well as auxiliary power demand are assumed constant, and vehicle mass may differ from the Tesla Model S because of the change of battery. These assumptions limit the accuracy of the range prediction but remain consistent with the engineering-oriented objectives of the study.

Overall, the proposed framework is well suited for battery pack sizing and comparative range analysis, and provides a solid baseline for future model refinements.

Part V

Credits

References

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